

$$\{X: X \subseteq \{3, 2, a\}, |X| = 2\}$$

$$\Rightarrow \underbrace{\emptyset, \{3\}, \{2\}, \{a\}, \{3, 2\}, \{3, a\}, \{2, a\}, \{3, 2, a\}}_{\substack{0 \quad 1 \quad 1 \quad 1 \quad 2 \quad 2 \quad 2 \quad 3 \\ \checkmark}}$$

$$\{\{3, 2\}, \{3, a\}, \{2, a\}\}$$

$$\{X: X \subseteq \{3, 2, a\}, |X| = 4\} = \emptyset = \{\}$$

$$\{X \subseteq \mathbb{N} : |X| \leq 1\}$$

$$\mathbb{N} = \{1, 2, 3, 4, 5, \dots, 99, \dots, 999, \dots, 1011, \dots\}$$

$$\mathcal{P}(\mathbb{N}) = \{\emptyset, \{1\}, \{2\}, \dots, \{99\}, \dots, \{1011\}, \dots\}$$

$$\{1, 2\}, \{1, 3\}, \dots, \{1, 99\}, \dots, \{1, 1011\}, \dots,$$

$$\{2, 3\}, \dots, \{2, 1011\}, \dots$$

$$\{3, 4\}, \dots$$

$$\{1, 2, 3, 4\}$$

$\vdots$

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$$\rightarrow \{\emptyset, \dots\} \leftarrow$$

$$\{\emptyset, \{1\}, \{2\}, \dots, \{99\}, \dots\}$$

$$\mathbb{R}^2 \subseteq \mathbb{R}^3?$$